

LATAM Bank

AI-first intake of unrecognized card charges, in Spanish and Portuguese

The model understands and writes. Code decides.

Live: <https://latam-bank-hackaton-2026.web.app> (online banking at /banca/, human expert console at /operador/)

Factored AI & Data Hackathon 2026. Synthetic data, fictional bank. All figures are offline measurements.

1. A problem chosen from the data

686,296

customer contacts from 148,443 customers, flat demand (+0.3%)

18.3%

of 67,095 complaints are unrecognized charges: the largest subcategory (12,297)

37 h

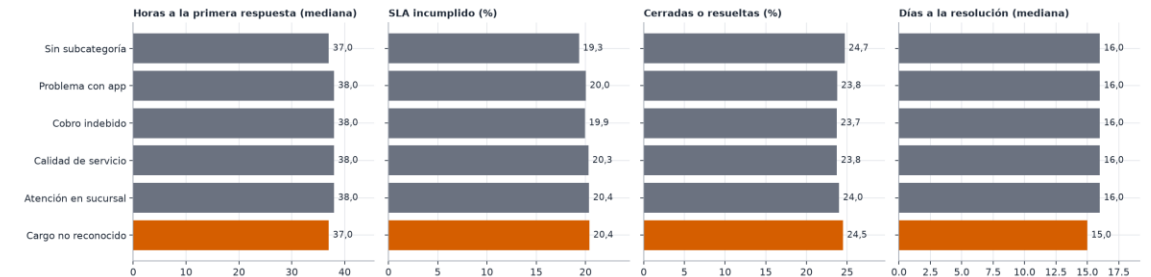
median first response, while fraud is contained in minutes

24.5%

closed or resolved; 20.4% breach the SLA

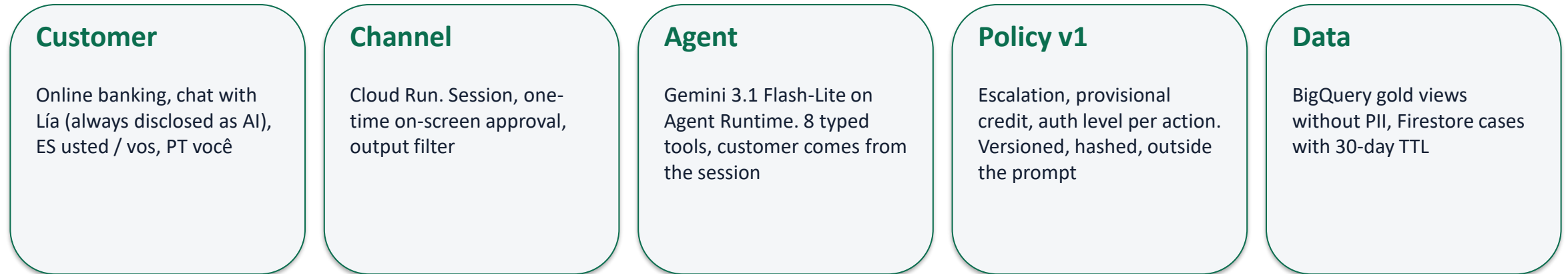
One focused workflow: transaction-dispute intake. It wins the prioritization score 4.00 vs 3.00 for account inquiries; the ranking flips only if volume weighs 40% or more, and we say so.

Línea base de quejas: el cargo no reconocido se atiende como las demás, y todas tardan



Data limits found and declared: 57% of amount_usd empty, 24% of digital events without customer, no Brazilian accounts.

2. Controlled automation: the LLM talks, policy-as-code acts



Normal path. The server pins the transaction, the agent reads it and proposes; nothing runs until the customer approves on screen. It reports only verified actions.

Ambiguous or unsupported. It lists candidates and asks; it abstains on credit and never invents eligibility. Policy questions are answered only from a retrieved source, with the rule cited, or it says it does not know.

Human intervention. A person on request, always. The expert console receives a 19-field handoff package (request, verified facts with source, actions done and not done, open questions, rule and version) and an AI copilot that drafts a reply the human edits and sends.

Unauthorized access and prompt injection are stopped in the tool layer, not in prose: other customers' data does not exist for the session.

3. Data engineering and ML with baselines, including a negative result

Data engineering

Bronze, silver, gold and platinum layers in BigQuery with dbt: 13 raw tables, 19M+ rows, 56 tests green.

ODCS contracts on every gold table, checked against live columns. Quality rules on bronze, chained SHA-256 manifest for lineage.

PII pseudonymized in silver with a key kept outside the repo; restricted attributes only for fairness audits.

Freshness policy and a labeled update fixture (10 cases: late arrival, redelivery, duplicates, bad currency).

Fraud score excluded from the agent: it leaks the label (2,373 of 2,373 above 30 are fraud).

Three learned components, each against a baseline

0.90

Policy retrieval
with citation (in
the workflow).
BM25: 0.52

0.40

Contact-reason
classifier, macro-
F1. Rules: 0.34

0.50

SLA-breach risk,
AUC. No signal: it
always abstains

Retrieval: 131 relevance judgments, stratified split, threshold chosen on validation, tested once on 68 questions. 0 citations for the 15 questions the policy does not cover.

Classifier: temporal split, leakage removed; it only helps after the call, and we report it as marginal.

Risk model: the label itself fails an audit; a permuted label performs the same. We ship the abstention, not a claim.

4. Measured quality and failures, on held-out cases

0 / 96

unsafe outcomes (upper bound 4%). Zero observed is not zero risk

29 / 36

safe automated resolution on resolvable cases (29/75 over all in scope)

1.4 s / 4.6 s

p50 / p95 per agent turn;
US\$0.002 per attempted case

15 / 15

injected failures end with no effects and no leaks

32 held-out cases, 3 runs each, Spanish and Portuguese: normal, ambiguous, unsupported, human-required, expired sessions, unauthorized access, direct and tool-borne prompt injection, BigQuery, Firestore and model outages. Deterministic verifiers, validated against 43 manually reviewed runs.

The honest headline: on that set the rules baseline passes 91% and the agent 80%, with overlapping intervals. The agent does not beat rules at following policy; it adds language, conversation and coverage that rules do not have.

The evaluation found real defects (a policy not enforced in a tool, an injected priority order obeyed, a list limit hiding charges, silent failures). All four were fixed in the tool layer and re-checked, reported as post hoc.

A new held-out set (43 cases, Mexico, Colombia and Argentina) was frozen by hash before building the latest capabilities, so the current system gets one clean measurement.

Fairness: outcomes compared by language, country and segment with small-sample flags; no attributable disparity in the historical baseline.

5. A credible route to operation, and what is still missing

In place and verified

Deployed end to end on Google Cloud: Firebase Hosting, Cloud Run, Agent Runtime, BigQuery, Firestore. 81 browser tests pass against production.

Tracing per turn, model call and tool, with no message content. Explanations rest on sources, rules and execution records, not on model reasoning.

Bounded retries, safe fallback to templates, human handoff that works without the model.

Least-privilege service account, secrets in Secret Manager, 30-day retention with Firestore TTL, uptime check and three alerts, Terraform planned with zero destroys.

588 unit tests, CI with data and credential guards.

Honest limits

All results are offline, on synthetic data, against a simulated customer from the same model family, with single-author labels.

No Brazilian accounts in the data: Portuguese is service in that language for regional customers, not reviewed by a native speaker.

One instance, sessions in memory, no load test. Voice and WhatsApp are designed, not deployed.

Banking tools are mocks with documented contracts; no money moves.

Before real operation: sessions in Firestore, load and cold-start measurement, individual identity with MFA for experts, a second labeler and a validated tone judge.